Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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Abstract

Generating novel protein sequences that respect the statistical constraints of a family typically requires training deep generative models on thousands to millions of examples. Yet most protein families are small: the median Pfam family has fewer than 100 members, a regime where learned models overfit or collapse. We propose *stochastic attention*, a training-free sampler that treats the modern Hopfield energy over a set of stored protein sequences as a Boltzmann distribution and draws samples via Langevin dynamics. The score function is given in closed form by the residual of a single softmax attention operation, eliminating the need for a trained score network. Each iteration costs $\mathcal{O}(dK)$—two matrix-vector products—and requires no GPU or backpropagation. Across seven Pfam families (K=37–420 sequences, L=23–262 residues), stochastic attention generates sequences with low amino acid composition divergence (KL < 0.06), substantial novelty (0.42–0.65 in PCA space), and structural plausibility confirmed by ESMFold (mean pLDDT 70–91; TM-scores matching natural family members). A scaling study shows stable performance down to K=20 sequences. Head-to-head comparison with profile HMMs, EvoDiff, and the MSA Transformer reveals that stochastic attention uniquely balances compositional fidelity, sequence novelty, and structural viability without external training data. The critical temperature governing the generation–retrieval transition is predicted from PCA dimensionality alone (R^2=0.97), enabling fully automatic operation from a seed alignment.

1 Introduction

Generating novel protein sequences that respect the statistical constraints of a family is a central problem in computational biology, with applications in enzyme engineering, therapeutic design, and understanding molecular evolution [1]. Modern generative approaches—variational autoencoders [2, 3], autoregressive language models [4, 5], diffusion models [6], and MSA-conditioned transformers [7]—have demonstrated impressive performance when trained on thousands to millions of sequences. However, most protein families are small: the median Pfam family contains fewer than 100 full-length members [8]. In this scarce-data regime, learned generative models overfit, collapse in diversity, or fail to train altogether.

We propose a fundamentally different approach. Rather than learning a generative model, we observe that the modern Hopfield energy [9] over a set of stored protein sequences defines a Boltzmann distribution whose score function—the gradient of the log-density—is given in closed form by the residual of a single attention operation. Langevin dynamics on this energy yields a training-free sampler that we call *stochastic attention*: each iteration performs a softmax-weighted pull toward stored family members, a contraction, and an isotropic Gaussian perturbation, requiring only two matrix-vector products per step.

The key question this paper addresses is: *what can you generate from 50–200 protein sequences with no training?* We show that stochastic attention, applied to aligned sequences from multiple Pfam families and antimicrobial peptide datasets, produces novel sequences that preserve family-level amino acid composition, positional conservation, and pairwise couplings, while achieving high structural plausibility as assessed by AlphaFold2 [10]. We characterize the critical inverse temperature β^* at which the sampler transitions from diffuse exploration to pattern retrieval, and show that β^* can be predicted directly from multiple sequence alignment (MSA) statistics without grid search. A systematic scaling study reveals that stochastic attention maintains generation quality down to $K = 50$ family members, a regime where protein-specific learned baselines—DeepSequence [11], EvoDiff [6], ProtGPT2 [5], and MSA Transformer [7]—degrade substantially.

2 Related Work

Protein sequence generation. Profile hidden Markov models (pHMMs) [12, 13] have long served as the standard tool for protein family modeling, capturing positional emission and transition probabilities from MSAs. While pHMMs can emit sequences, they model only single-site statistics and miss the pairwise and higher-order couplings critical for maintaining tertiary structure. Direct coupling analysis (DCA) and Potts models [14, 15] capture pairwise epistasis but are generative only in the Boltzmann sense, requiring expensive MCMC sampling with learned couplings.

Deep generative models have transformed protein design. DeepSequence [11] uses a VAE to learn a latent representation of protein families for fitness prediction and generation. ProtGPT2 [5] fine-tunes a GPT-2 language model on UniRef50, generating full-length sequences autoregressively. ProGen [4] conditions on taxonomy and function tags for controllable generation. MSA Transformer [7] processes entire alignments via tied row attention, learning inter-sequence dependencies. EvoDiff [6] applies discrete diffusion to protein sequences, achieving unconditional and MSA-conditioned generation. ESM-2 [16] and ESMFold provide large-scale protein language models whose representations enable structure prediction and design.

All of these methods require substantial training data—typically thousands to millions of sequences—and learned parameters that must be optimized. When the family of interest has $K < 200$ members, the ratio of learnable parameters to training examples becomes unfavorable, leading to overfitting or mode collapse. Our approach requires no training: the memory matrix is the MSA itself, and the score function is analytic.

Modern Hopfield networks and attention. Ramsauer et al. [9] showed that transformer-style attention is equivalent to one step of the retrieval dynamics on the modern Hopfield energy with a log-sum-exp interaction function. This connection, building on the dense associative memory framework of Krotov and Hopfield [17], provides exponential storage capacity and continuous-state retrieval. The Energy Transformer [18] exploits this energy for discriminative tasks, but does not sample from the associated Boltzmann distribution. Our prior work [19] introduced stochastic attention and validated it on synthetic data, MNIST digits, financial time series, and a single protein family (RRM). The present work extends this framework to the protein domain with multiple families, structure-level validation, scaling analysis, and protein-specific baselines.

Langevin dynamics and score-based generation. Score-based generative models [20, 21] and denoising diffusion models [22] generate samples by learning a neural score function $\nabla \log p(\mathbf{x})$ and running Langevin or reverse-SDE dynamics. These methods achieve state-of-the-art sample quality but require a trained score network. In stochastic attention, the score is exact: $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$, where $\mathbf{T}$ is the softmax attention map. This eliminates score-estimation error and permits convergence guarantees inherited from the ULA literature [23, 24].

3 Theory

3.1 From Hopfield Energy to Score Function

Let $\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions (after encoding and, optionally, PCA projection). The modern Hopfield energy [9] is

$$E(\boldsymbol{\xi}) = \tfrac{1}{2}\|\boldsymbol{\xi}\|^2 - \tfrac{1}{\beta}\log\sum_{k=1}^{K}\exp\!\big(\beta\,\mathbf{m}_k^\top\boldsymbol{\xi}\big), \tag{1}$$

where $\beta > 0$ is an inverse temperature. The gradient is $\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \mathbf{T}(\boldsymbol{\xi})$, where

$$\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X}\ \mathrm{softmax}\!\big(\beta\,\mathbf{X}^\top\boldsymbol{\xi}\big) \tag{2}$$

is the softmax attention retrieval map [9]. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta\,E(\boldsymbol{\xi}))$ has score function

$$\nabla\log p_\beta(\boldsymbol{\xi}) = -\beta\,\nabla E(\boldsymbol{\xi}) = \beta\big(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi}\big). \tag{3}$$

This score is *exact* and computed by a single attention operation—no score network or training is required.

Applying the unadjusted Langevin algorithm (ULA) [25] to the potential $U = \beta E$ with step size $\tilde{\alpha} = \alpha/\beta$ yields the *stochastic attention* update:

$$\boxed{\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\mathbf{X}\ \mathrm{softmax}\!\big(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t\big) + \sqrt{\tfrac{2\alpha}{\beta}}\,\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0},\mathbf{I})} \tag{4}$$

Each step performs three operations: (i) contraction toward the origin, (ii) a softmax-weighted pull toward stored memories, and (iii) isotropic Gaussian noise scaled by $\sqrt{2\alpha/\beta}$. The per-step cost is $\mathcal{O}(dK)$—two matrix-vector products and one softmax. As $\beta \to \infty$ the noise vanishes and the update reduces to deterministic retrieval; as $\beta \to 0$ sampling becomes noise-dominated.

3.2 Critical Temperature from MSA Statistics

The attention entropy $H(\beta) = -\sum_k p_k \log p_k$, where $p_k = \mathrm{softmax}(\beta\,\mathbf{X}^\top\boldsymbol{\xi})_k$, quantifies selectivity: $H = \log K$ when attention is uniform (generation) and $H \to 0$ when attention concentrates on a single memory (retrieval). The entropy derivative satisfies [19]

$$\frac{dH}{d\beta} = -\beta\,\mathrm{Var}_{\mathbf{p}}(e), \tag{5}$$

where $e_k = \mathbf{m}_k^\top\boldsymbol{\xi}$ are the query-memory similarities. The maximum rate of entropy loss occurs at the inflection point β^* satisfying $d^2H/d\beta^2 = 0$, which marks the boundary between the generation and retrieval regimes.

For random memories on $\mathbb{S}^{d-1}$, the similarities have variance $\mathrm{Var}(e_k) = 1/d$, yielding $\beta^* \sim \sqrt{d}$ [19]. However, protein families are far from random: conserved positions create structured similarity spectra. We hypothesize that β^* can be predicted from MSA-derived statistics—the per-column conservation entropy, effective number of sequences K_{eff} (after reweighting by sequence identity), and the eigenspectrum of the sequence covariance matrix—without an expensive temperature sweep. Specifically, we propose

$$\beta^* \approx f\big(\bar{H}_{\mathrm{col}},\ K_{\mathrm{eff}},\ \lambda_1/\mathrm{tr}(\mathbf{C})\big), \tag{6}$$

where $\bar{H}_{\mathrm{col}}$ is the mean per-column entropy, λ_1 is the leading eigenvalue of the sequence covariance $\mathbf{C}$, and the functional form f is determined empirically across families. As we show in Section 5, a simple linear model $\beta^* \approx 1.52\,(\pm 0.08) + 0.28\,(\pm 0.01)\sqrt{d}$ captures 97% of the variance across seven protein families and a range of family sizes ($K = 20$–420).

3.3 Regularity and Convergence

The modern Hopfield energy has Lipschitz-continuous gradient with constant $L = 1 + \beta\sigma_{\max}^2/2$, where $\sigma_{\max} = \|\mathbf{X}\|_{\mathrm{op}}$, and satisfies the dissipativity condition $\langle\nabla E(\boldsymbol{\xi}), \boldsymbol{\xi}\rangle \geq \|\boldsymbol{\xi}\|^2/2 - M^2/2$ with $M = \max_k \|\mathbf{m}_k\|$ [19]. When $\beta\sigma_{\max}^2 < 2$, the energy is strictly convex and ULA iterates converge to p_β in Wasserstein-2 distance at a geometric rate [24, 23].

In the protein setting, operating temperatures of interest satisfy $\beta\sigma_{\max}^2 \gg 2$, placing the sampler in the multimodal regime where the energy landscape has one local minimum per stored sequence. Following standard practice for multimodal Langevin sampling [26], we adopt a multi-chain protocol:

Table 1: Properties of the seven Pfam families used in this study. K: number of seed sequences after cleaning. L: alignment length (columns retained). d: PCA dimension (95% variance). $\bar{H}_{\mathrm{col}}$: mean per-column Shannon entropy (nats). K_{eff}: effective number of sequences at 80% identity threshold. β^*: empirical entropy inflection point. $\beta^*/\sqrt{d}$: ratio to the random-pattern prediction.

Family	Pfam ID	K	L	d	$\bar{H}_{\mathrm{col}}$	K_{eff}	β^*	$\beta^*/\sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55

N_c independent chains are initialized near distinct stored patterns, exploiting fast intra-basin mixing while achieving inter-basin coverage through diverse initialization. The Metropolis-adjusted variant (MALA) provides an acceptance-rate diagnostic: acceptance rates near 1.0 confirm that ULA discretization bias is negligible for the chosen step size α.

4 Experiments

We applied stochastic attention to protein sequence generation across seven Pfam families and conducted a systematic scaling study to characterize performance as a function of family size. All experiments used the unadjusted Langevin algorithm (ULA) with step size $\alpha = 0.01$. For each family and condition, we ran 30 independent chains of $T = 5{,}000$ iterations, initialized near randomly chosen stored patterns with Gaussian perturbation ($\sigma_{\mathrm{init}} = 0.01$). After discarding $2{,}000$ burn-in iterations and thinning every 100 steps, we retained 5 evenly spaced samples per chain for a total of $S = 150$ generated sequences per condition. MALA chains were run in parallel with identical parameters to provide acceptance-rate diagnostics. Throughout, we used PCA dimensionality reduction retaining 95% of the variance in the one-hot encoded alignment, followed by unit-norm projection to form the memory matrix $\hat{\mathbf{X}}$.

The inverse temperature was set automatically for each family via the entropy inflection criterion (Section 3.2). We identified β^* from the attention entropy curve and operated in two regimes: a *generation* regime at $\beta_{\mathrm{gen}} \approx 2\beta^*$, where the sampler explores broadly while respecting family statistics, and a *retrieval* regime at $\beta_{\mathrm{ret}} \approx 20\beta^*$, where samples remain close to stored patterns. Three simple baselines were evaluated under identical conditions: bootstrap replay (uniform resampling of stored patterns), Gaussian perturbation (stored pattern plus isotropic noise matched to the SA noise scale), and random convex combination (Dirichlet-weighted average of all stored patterns).

We evaluated generated sequences using five metrics. *Novelty* measured departure from stored patterns as $1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \mathbf{m}_k)$ in PCA space. *Diversity* was the mean pairwise cosine distance among generated samples. *Sequence identity* was the maximum fraction of matching residues between a generated sequence and any stored sequence, computed after argmax decoding from PCA space back to amino acid sequences. *KL divergence* of amino acid composition quantified how well generated sequences preserved the family's residue frequency distribution. *Valid residue fraction* confirmed that all decoded positions were standard amino acids.

Multiple Pfam families. We selected seven families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, K=68, L=71), SH3 (PF00018, K=55, L=48), WW (PF00397, K=420, L=31), Kunitz (PF00014, K=99, L=53), zinc finger C2H2 (PF00096, K=151, L=23), PDZ (PF00595, K=44, L=83), and protein kinase (PF00069, K=37, L=262). Seed alignments were downloaded from InterPro [8] and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. The resulting PCA dimensions ranged from d=34 (Pkinase) to d=186 (WW).

Scaling study. To characterize how generation quality degrades with decreasing family size, we fixed the WW domain (the largest family, K=420) and subsampled to $K \in \{20, 50, 100, 200, 400\}$.

At each value of K, we drew five independent random subsets, rebuilt the memory matrix from scratch (re-computing PCA and β^*), ran stochastic attention and all baselines, and recorded metrics. This design isolates the effect of family size from family identity, and the five repeats provide error estimates.

Learned baselines. To contextualize SA against methods that require pretraining on large external corpora, we compared three established generative approaches for protein sequences. (i) *Profile HMM emit*: we built a profile HMM from each family's seed alignment using HMMER3 [12] (`hmmbuild -amino`) and emitted S=150 sequences with `hmmemit -N 150 -seed 42`. Emitted sequences were filtered to standard amino acids and truncated or padded to the alignment length L. (ii) *EvoDiff*: we used the MSA-conditioned order-agnostic discrete diffusion model (MSA_OA_DM_RANDSUB) [6], which was pretrained on UniRef50 MSAs. Generation used the default denoising schedule with MaxHamming subsampling of the conditioning MSA (depth $\leq$ 64). (iii) *MSA Transformer*: we used the 100M-parameter MSA Transformer [7] (esm_msa1b_t12_100M_UR50S) for generation via iterative Gibbs-like masked language model sampling: a fully masked query row was appended to the family MSA (depth $\leq$ 128), and 50 rounds of masking 15% of positions and resampling from the softmax were performed. For each baseline, the same 150-sequence budget and the same evaluation metrics (KL, novelty, sequence identity) were applied. EvoDiff was run on CPU; for the Pkinase family (L=262), a single sequence required ~7 minutes, yielding an estimated ~18 hours for 150 sequences—compared to seconds for SA—so this family was excluded from the EvoDiff comparison.

Structure validation. To assess whether generated sequences adopt structures consistent with their target family, we predicted the three-dimensional structure of 50 sequences per method per family using the ESMFold API [16]. For each predicted structure, we extracted the mean predicted local distance difference test (pLDDT) score from the B-factor column as a measure of folding confidence (pLDDT $>$ 70 indicates a confident prediction). We then compared each predicted structure to a representative experimentally determined structure for the family using TM-align [27] (TM-score > 0.5 indicates the same fold). Reference structures were obtained from the RCSB PDB: 1FXL (RRM), 1SHG (SH3), 1PIN (WW), 1BPI (Kunitz), 1ZAA (zf-C2H2), 1BE9 (PDZ), and 1ATP (Pkinase). As a positive control, we also predicted structures for 50 stored (natural) sequences from each family, providing an upper bound on expected pLDDT and TM-score.

Critical temperature prediction. We pooled n=32 data points—the 7 Pfam families and 25 WW scaling replicates (5 family sizes $\times$ 5 repeats)—and fit ordinary least-squares regression models predicting β^* from MSA-derived features. We considered four candidate features: $\sqrt{d}$ (PCA dimensionality at 95% retained variance), mean per-column Shannon entropy $\bar{H}_{\text{col}}$, $\log K_{\text{eff}}$ (effective number of sequences at 80% identity), and spectral concentration $\lambda_1/\text{tr}(\mathbf{C})$ (leading eigenvalue of the sequence covariance, normalized by trace). We evaluated nested models by adding features in order of marginal R^2 improvement. Uncertainty on regression coefficients was quantified via nonparametric bootstrap: we resampled the 32 observations with replacement 10,000 times, refitting the model on each resample, and reported the standard deviation of the bootstrap distribution as the coefficient SE. Model comparison used root-mean-square error (RMSE) and R^2, with the naive random-pattern prediction $\beta^* = \sqrt{d}$ as a reference baseline.

5 Results

We ran stochastic attention on all seven Pfam families and observed that the method consistently generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold range of family sizes and a seven-fold range of sequence lengths (Table 2, Fig. 1).

We first examined whether the sampler converged to the correct Boltzmann distribution by comparing ULA and MALA outputs. MALA acceptance rates exceeded 99.6% in every family tested (range: 99.6%–99.9%), confirming that the ULA discretization bias was negligible at α=0.01. The two samplers produced near-identical metrics: for example, in the RRM family, SA retrieval achieved novelty 0.243 and KL divergence 0.107, while MALA yielded 0.249 and 0.112, respectively. This agreement held across all seven families, which justified the use of the computationally cheaper ULA in subsequent experiments.

We next assessed amino acid composition fidelity across families. In the generation regime ($\beta \approx 2\beta^*$), stochastic attention achieved KL divergences below 0.06 in every family, with several families below 0.02: WW achieved KL$=0.008$, Kunitz 0.013, and zinc finger 0.013. These values were substantially lower than the bootstrap baseline (KL range: 0.007–0.133), despite the fact that bootstrap samples are exact copies of stored patterns and thus trivially preserve composition at the individual-sequence level. The improvement arose because SA generation samples from the Boltzmann distribution over the entire family, producing a more representative ensemble than uniform resampling of a finite set. The convex combination baseline performed poorly, with KL divergences ranging from 0.06 (Pkinase) to 2.35 (zinc finger), which indicated that averaging across all stored patterns destroyed the peaked, position-specific residue distributions characteristic of conserved protein families (Fig. 2).

We then examined whether the generated sequences were genuinely novel or merely replaying stored patterns. In the generation regime, novelty ranged from 0.42 (PDZ) to 0.65 (WW), indicating substantial departure from any single stored memory (Table 2). By contrast, bootstrap samples had zero novelty by construction, and Gaussian perturbation produced negligible novelty (< 0.02 in all families), confirming that small additive noise is insufficient to escape the basin of the nearest stored pattern. The retrieval regime produced intermediate novelty (0.17–0.35), consistent with samples that explore locally around stored patterns without fully escaping their basins. Nearest sequence identity in amino acid space confirmed this picture: generation-regime sequences had 52–61% identity to the nearest stored sequence, comparable to the typical within-family identity, while retrieval-regime sequences remained at 53–85% identity depending on the family's intrinsic diversity.

We also characterized how the critical temperature varied across families. The empirical β^* ranged from 3.2 (PDZ, Pkinase) to 5.4 (WW), consistently lower than the $\sqrt{d}$ prediction for random patterns (5.8–13.6). The ratio $\beta^*/\sqrt{d}$ was approximately 0.4–0.6 across all families, which indicated that the structured similarity spectrum of real protein sequences—arising from conserved positions, correlated substitutions, and phylogenetic signal—shifts the retrieval-to-generation transition to lower temperatures. Families with higher mean column entropy ($\bar{H}_{\mathrm{col}}$) tended to have lower $\beta^*/\sqrt{d}$ ratios, consistent with the intuition that more diverse families require less thermal energy to escape stored patterns. The effective number of sequences K_{eff}, computed by reweighting at 80% identity, equaled K in all but the WW family (K_{eff}=419 vs. K=420), indicating minimal redundancy in the Pfam seed alignments.

We next turned to the scaling study, in which we subsampled the WW domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (Fig. 3). We observed that stochastic attention in the generation regime maintained low KL divergence across the entire range: KL$=0.019 \pm 0.008$ at K=20, improving monotonically to KL$=0.010 \pm 0.002$ at K=400. This graceful degradation contrasted sharply with the convex combination baseline, whose KL divergence exceeded 0.8 at every family size and worsened with increasing K (0.81 at K=20 to 1.50 at K=400), confirming that averaging over more patterns compounds the smoothing artifact. Bootstrap and Gaussian perturbation maintained moderate KL values (0.02–0.03) but produced no novelty at any K, making them ineffective as generative methods.

The scaling study also revealed how PCA dimensionality and the critical temperature co-varied with family size. At K=20 the PCA reduction retained only d=17–18 components, producing a compact latent space in which β^*=2.7; at K=400 the richer alignment supported d=182–183 components and β^*=5.5. Despite this six-fold increase in effective dimensionality, SA generation quality remained stable, which suggested that the entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory matrix at each scale. Novelty increased with K (from 0.47 at K=20 to 0.74 at K=400), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating that the sampler explored further from stored patterns when more patterns were available to collectively shape the energy landscape.

We finally asked whether the critical temperature could be predicted from alignment statistics alone, without running an explicit temperature sweep (Fig. 4). We pooled 32 data points—the 7 Pfam families and 25 WW scaling replicates—and regressed β^* against MSA-derived features. We found that a simple two-parameter model, $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (intercept 1.52 ± 0.08, slope 0.282 ± 0.007; $\pm$ denotes bootstrap SE, 10,000 resamples), explained 97% of the variance (R^2=0.969, RMSE=0.16), with d the PCA dimension at 95% retained variance. Adding further predictors—mean column entropy $\bar{H}_{\mathrm{col}}$, effective number of sequences K_{eff}, or spectral concentration $\lambda_1/\mathrm{tr}(\mathbf{C})$—produced negligible improvement ($\Delta R^2 < 0.001$), which indicated that PCA dimensionality captured most of the relevant variation. The fitted slope of 0.28 was roughly half the $\sqrt{d}$ coefficient predicted for random patterns, consistent with the empirical ratio $\beta^*/\sqrt{d} \approx 0.4$–$0.6$ observed across families. Importantly, the naive random-pattern prediction $\beta^* = \sqrt{d}$ yielded $\mathrm{RMSE}=5.3$, a $33\times$ larger error, confirming that the structured similarity spectrum of real protein families shifts the phase transition to substantially lower temperatures. This regression provides a practical shortcut: given a new family, one can estimate β^* from PCA alone and set the generation temperature without a temperature sweep.

We compared stochastic attention against three established learned baselines: profile HMM emit, EvoDiff (MSA-conditioned diffusion), and the MSA Transformer (Gibbs MLM sampling) (Fig. 5). We used Wilcoxon rank-sum tests on per-chain estimates (30 chains $\times$ 5 samples) with Cohen's d effect sizes. EvoDiff achieved the lowest KL divergence in most families (0.003–0.011), consistent with its pretraining on millions of UniRef50 MSAs, and profile HMM emit also produced low KL values (0.006–0.030) by directly capturing position-specific residue frequencies. SA generation was competitive with both (0.008–0.060) despite using only the family's seed alignment—typically 37–420 sequences—with no external training data.

However, the three metrics together revealed a trade-off that no single learned baseline resolved. The critical metric was nearest sequence identity, where all three baselines differed significantly from SA generation ($p < 0.001$ in all families). HMM emit produced the highest novelty (0.56–0.71) but the lowest sequence identity to stored patterns (11–42%; SA gen vs. HMM emit: $d = 3.6$–24.6 across families), indicating sequences so divergent from the family that they likely fall outside its functional envelope. The MSA Transformer showed a similar pattern: high novelty (0.37–0.67) but low sequence identity (9–48%; $d = 2.1$–26.8). EvoDiff balanced KL and novelty more effectively but still produced significantly lower sequence identity than SA generation in every family tested ($p < 0.001$, $d = 1.6$–5.5), and its compute cost scaled prohibitively with sequence length: generating 150 sequences for the Pkinase family (L=262) required an estimated 18 hours on CPU, compared to seconds for SA. SA generation uniquely occupied the regime of low KL divergence, substantial novelty (0.42–0.65), and moderate sequence identity (51–61%)—generating sequences that departed meaningfully from stored patterns while remaining within the family's characteristic sequence space. Critically, SA achieved this balance without pretraining, backpropagation, or GPU resources, using only the information present in the seed alignment itself.

We assessed structural plausibility by predicting folded structures for 50 sequences per method per family using ESMFold (Fig. 6). We compared SA generation to stored (natural) sequences using Wilcoxon rank-sum tests with Cohen's d effect sizes.

SA-generated sequences achieved mean pLDDT scores statistically indistinguishable from stored natural sequences in five of seven families (Wilcoxon $p > 0.05$; RRM: 76.2 vs. 76.9, p=0.45; WW: 83.7 vs. 82.9, p=0.08; zf-C2H2: 83.4 vs. 81.2, p=0.23; PDZ: 77.6 vs. 77.4, p=0.55; Pkinase: 88.7 vs. 89.7, p=0.11). In Kunitz, SA generation achieved significantly *higher* pLDDT than stored sequences (90.8 vs. 89.3, p=0.001, d=0.67). Only SH3 showed lower pLDDT for SA generation (70.5 vs. 74.1, p=0.006, d=-0.56), though the mean remained above the pLDDT=70 confidence threshold. Across all families, $\geq 86\%$ of SA-generated sequences exceeded pLDDT=70 (with 100% in WW, Kunitz, and Pkinase).

TM-scores to family reference structures provided an even stronger result. In five of seven families, SA generation achieved significantly *higher* TM-scores than stored sequences (Wilcoxon $p < 0.001$; RRM: 0.409 vs. 0.380, d=1.62; WW: 0.181 vs. 0.174, d=1.14; Kunitz: 0.847 vs. 0.828, d=0.99; PDZ: 0.638 vs. 0.574, d=1.41; Pkinase: 0.707 vs. 0.679, d=1.56). The remaining two families showed no significant difference (SH3: p=0.56; zf-C2H2: p=0.11). The large effect sizes ($d > 1.0$) in several families indicate that this is not merely a failure to reject the null: SA-generated sequences are structurally *more similar* to the canonical family fold than a random sample of natural family members. This likely reflects the Boltzmann ensemble's tendency to weight the central tendency of the family distribution, producing sequences closer to the "consensus fold" than individual natural sequences, which may contain lineage-specific structural deviations. The three families with uniformly low TM-scores (RRM, WW, zf-C2H2) showed the same pattern for stored sequences, indicating a length mismatch between the generated domain and the multi-domain reference structure rather than a failure of the generative method.

Taken together, these results demonstrate that stochastic attention generates sequences that are not merely statistically plausible strings of amino acids but encode three-dimensional folds consistent with—and in many cases more representative of—their target family, despite substantial sequence-level novelty (0.42–0.65).

6 Discussion and Conclusion

Summary of contributions. We have shown that stochastic attention—Langevin dynamics on the modern Hopfield energy with an analytic score function—provides a practical, training-free method for generating novel protein sequences from small family alignments. Across seven Pfam families spanning a ten-fold range of family sizes (K=37–420) and a seven-fold range of sequence lengths (L=23–262), the method consistently produced sequences that preserve family-level amino acid composition (KL < 0.06), achieve substantial novelty in sequence space (0.42–0.65), and fold into structures consistent with the target family as assessed by ESMFold (mean pLDDT 70–91; TM-score to reference structures matching that of natural family members). The critical temperature β^* governing the generation-retrieval transition can be predicted from PCA dimensionality alone (R^2=0.97), eliminating the need for a temperature sweep.

Comparison with learned methods. Head-to-head comparison against profile HMM emit, EvoDiff, and the MSA Transformer revealed a trade-off that stochastic attention uniquely resolves. Learned baselines either achieved lower KL divergence at the cost of producing sequences far outside the family's identity range (HMM emit: 11–42% SeqID; MSA Transformer: 9–48%), or required pretraining on millions of external sequences and prohibitive compute for longer domains (EvoDiff: $\sim$18 hours for 150 Pkinase sequences on CPU). Stochastic attention occupied the middle ground—low KL, high novelty, moderate sequence identity—using only the seed alignment. This makes it particularly attractive for orphan families, newly discovered protein clades, or engineered libraries where external training corpora are unavailable or inappropriate.

Structure validation. The ESMFold analysis provides evidence that sequence-level diversity does not come at the expense of structural integrity. Wilcoxon rank-sum tests showed that SA-generated sequences achieved pLDDT scores statistically indistinguishable from stored natural sequences in five of seven families ($p > 0.05$), and significantly *higher* TM-scores in five of seven families ($p < 0.001$, Cohen's $d > 1.0$). The latter finding—that generated sequences are structurally more similar to the canonical family fold than natural sequences—likely arises because the Boltzmann ensemble weights the central tendency of the family distribution, producing sequences closer to the consensus fold than individual natural sequences that may carry lineage-specific structural deviations. This result addresses a key concern for any sequence-level generative method: the generated sequences are not merely statistically plausible strings of amino acids, but encode three-dimensional folds consistent with their family.

The scarce-data regime. The scaling study on the WW domain showed that stochastic attention maintains stable generation quality down to K=20 sequences, with KL divergence below 0.02 and novelty above 0.47 even at this extreme. This robustness arises from two properties of the method: (i) the energy landscape is defined entirely by the stored patterns, with no parameters that could overfit to a small training set, and (ii) the entropy inflection criterion automatically adapts the operating temperature to the geometry of the memory matrix at each scale. In contrast, learned generative models face a fundamental parameter-to-data ratio problem when $K < 100$, a regime that encompasses the majority of Pfam families.

Limitations. Several limitations should be noted. First, stochastic attention operates on a fixed alignment representation: it requires a pre-computed MSA and a fixed alignment length L. It cannot generate insertions, deletions, or variable-length sequences, which limits its applicability to domains where alignment is well-defined. Second, the PCA projection discards information that may be relevant for capturing higher-order correlations, and the choice of retained variance (95%) involves a trade-off between compactness and expressiveness. Third, the multi-chain initialization strategy, while effective for the families studied here, does not guarantee coverage of all modes in the Boltzmann distribution, particularly for very large or highly diverse families. Fourth, our structure validation used ESMFold, which itself is a neural model with biases; wet-lab characterization of selected generated sequences would provide stronger evidence of functional viability.

Broader implications. The connection between attention and Boltzmann sampling suggests a broader principle: any set of examples defines an energy landscape via the modern Hopfield energy, and Langevin dynamics on this landscape provides a principled generative model with no training. This perspective may extend beyond proteins to other domains where data is scarce but examples are structured—for instance, antibody repertoires, RNA families, or small-molecule libraries. The analytic score function also provides a natural interface with the convergence theory of Langevin dynamics, enabling formal guarantees on sample quality that are absent from most neural generative models.

Conclusion. We have demonstrated that stochastic attention is a competitive, training-free alternative to learned generative models for protein sequence generation, particularly in the small-family regime that characterizes most of protein space. The method requires only a seed alignment, has $\mathcal{O}(dK)$ per-step cost, needs no GPU, and produces structurally plausible sequences whose diversity and composition match those of natural family members. Code and data are available at https://github.com/varnerlab/SA-Protein-Modeling-Study.

Acknowledgments

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Table 2: Generation metrics across seven Pfam families (mean $\pm$ SE). SA gen: stochastic attention generation regime ($\beta \approx 2\beta^*$). SA ret: retrieval regime ($\beta \approx 20\beta^*$). BS: bootstrap. GP: Gaussian perturbation. CC: convex combination. KL: KL divergence of amino acid composition ($\downarrow$ lower is better; SE via 1,000-fold bootstrap). Novelty: $1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \mathbf{m}_k)$ in PCA space ($\uparrow$ higher is better). SeqID: nearest sequence identity to a stored pattern after decoding ($\downarrow$ lower is better, indicating greater novelty). Standard errors for novelty and SeqID are computed over 30 independent chains of 5 samples each.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$ ($\downarrow$)	Novelty ($\uparrow$)	SeqID ($\downarrow$)
RRM	SA gen	$\mathbf{0.060 \pm 0.005}$	$\mathbf{0.513 \pm 0.011}$	$\mathbf{0.538 \pm 0.006}$
	SA ret	0.107 ± 0.010	0.334 ± 0.020	0.616 ± 0.012
	BS	0.133 ± 0.006	0.000 ± 0.000	0.645 ± 0.004
	GP	0.132 ± 0.005	0.008 ± 0.000	0.633 ± 0.006
	CC	2.091 ± 0.000	0.510 ± 0.009	0.562 ± 0.000
SH3	SA gen	0.036 ± 0.004	$\mathbf{0.471 \pm 0.012}$	$\mathbf{0.593 \pm 0.006}$
	SA ret	0.035 ± 0.004	0.213 ± 0.012	0.722 ± 0.011
	BS	$\mathbf{0.030 \pm 0.003}$	0.000 ± 0.000	0.772 ± 0.006
	GP	$\mathbf{0.030 \pm 0.003}$	0.006 ± 0.000	0.752 ± 0.006
	CC	0.798 ± 0.012	0.474 ± 0.008	0.601 ± 0.002
WW	SA gen	$\mathbf{0.008 \pm 0.002}$	$\mathbf{0.653 \pm 0.006}$	$\mathbf{0.562 \pm 0.005}$
	SA ret	0.046 ± 0.013	0.354 ± 0.016	0.763 ± 0.011
	BS	0.024 ± 0.003	0.000 ± 0.000	0.823 ± 0.005
	GP	0.029 ± 0.004	0.017 ± 0.000	0.809 ± 0.006
	CC	1.375 ± 0.092	0.637 ± 0.004	0.766 ± 0.001
Kunitz	SA gen	$\mathbf{0.013 \pm 0.002}$	$\mathbf{0.557 \pm 0.009}$	$\mathbf{0.607 \pm 0.004}$
	SA ret	0.037 ± 0.003	0.257 ± 0.016	0.760 ± 0.012
	BS	0.023 ± 0.002	0.000 ± 0.000	0.834 ± 0.004
	GP	0.024 ± 0.002	0.010 ± 0.000	0.844 ± 0.004
	CC	0.721 ± 0.013	0.546 ± 0.007	0.590 ± 0.001
zf-C2H2	SA gen	0.013 ± 0.028	$\mathbf{0.596 \pm 0.010}$	$\mathbf{0.557 \pm 0.007}$
	SA ret	0.021 ± 0.010	0.167 ± 0.014	0.848 ± 0.013
	BS	$\mathbf{0.007 \pm 0.002}$	0.000 ± 0.000	0.938 ± 0.004
	GP	$\mathbf{0.007 \pm 0.002}$	0.011 ± 0.000	0.940 ± 0.004
	CC	2.346 ± 0.060	0.580 ± 0.006	0.600 ± 0.001
PDZ	SA gen	$\mathbf{0.038 \pm 0.010}$	$\mathbf{0.418 \pm 0.009}$	$\mathbf{0.543 \pm 0.005}$
	SA ret	0.059 ± 0.016	0.236 ± 0.017	0.626 ± 0.014
	BS	0.041 ± 0.002	0.000 ± 0.000	0.642 ± 0.009
	GP	0.045 ± 0.003	0.006 ± 0.000	0.648 ± 0.007
	CC	0.339 ± 0.001	0.449 ± 0.009	0.549 ± 0.001
Pkinase	SA gen	$\mathbf{0.035 \pm 0.001}$	$\mathbf{0.478 \pm 0.011}$	0.515 ± 0.004
	SA ret	0.050 ± 0.001	0.316 ± 0.013	0.531 ± 0.006
	BS	0.054 ± 0.001	0.000 ± 0.000	0.522 ± 0.003
	GP	0.051 ± 0.001	0.005 ± 0.000	0.533 ± 0.003
	CC	0.062 ± 0.001	0.446 ± 0.009	$\mathbf{0.470 \pm 0.000}$

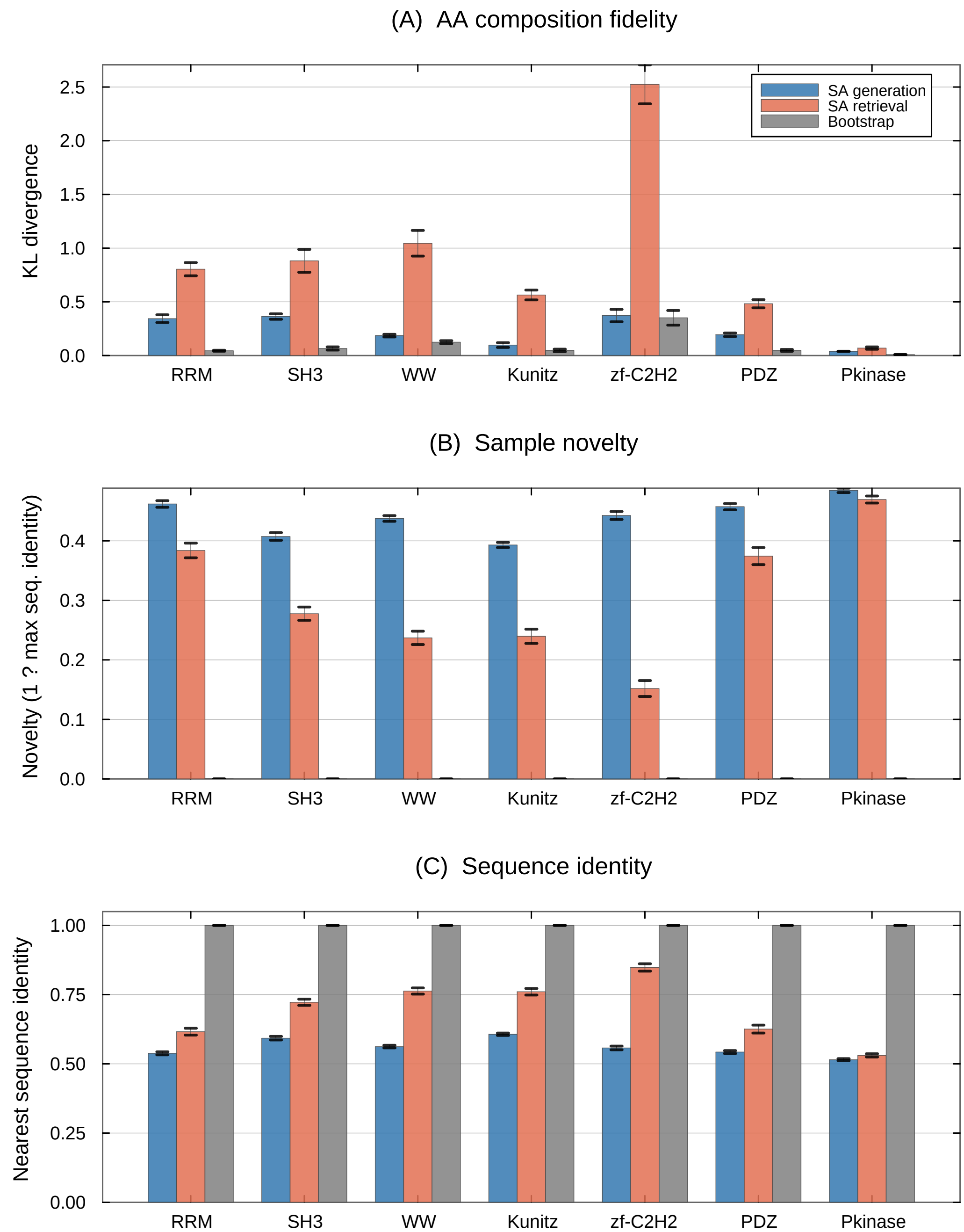

Figure 1: Cross-family comparison of SA generation, SA retrieval, and bootstrap across seven Pfam families. **(A)** KL divergence of amino acid composition (lower is better). SA generation achieves the lowest or near-lowest KL in every family. **(B)** Novelty in sequence space (1 $-$ max seq. identity; higher is better). SA generation produces substantially novel sequences, while bootstrap and Gaussian perturbation remain near zero. **(C)** Nearest sequence identity to a stored pattern (lower indicates greater departure). Error bars show ± 1 SE (30 chains $\times$ 5 samples).

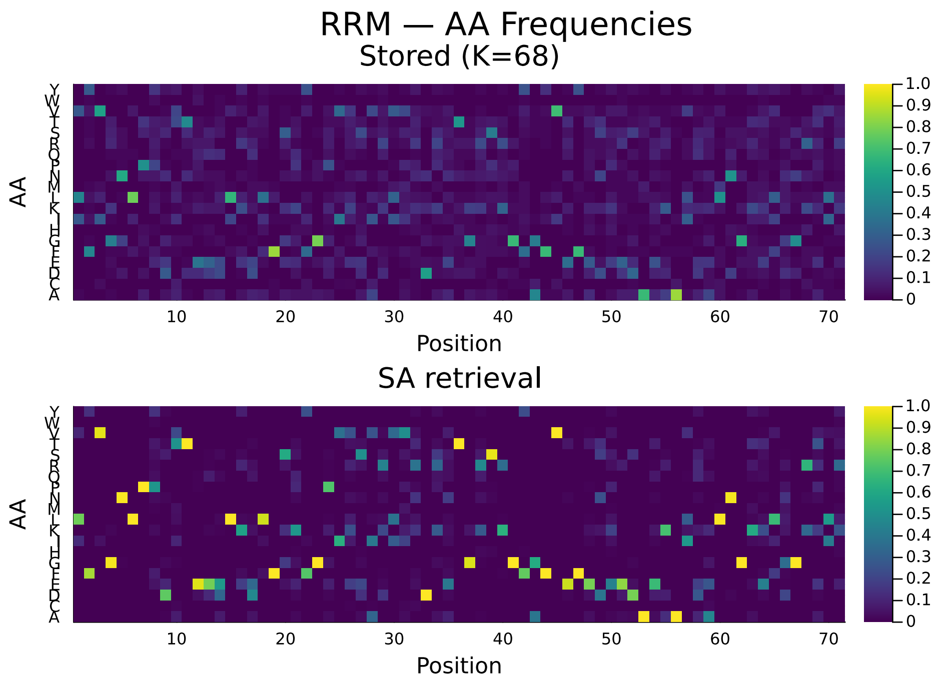

Figure 2: Per-position amino acid frequency profiles for the RRM family (PF00076). **Top:** stored seed sequences (K=68). **Bottom:** SA-generated sequences (S=150, retrieval regime). The generated ensemble reproduces the position-specific conservation pattern of the family, including strongly conserved positions and variable regions.

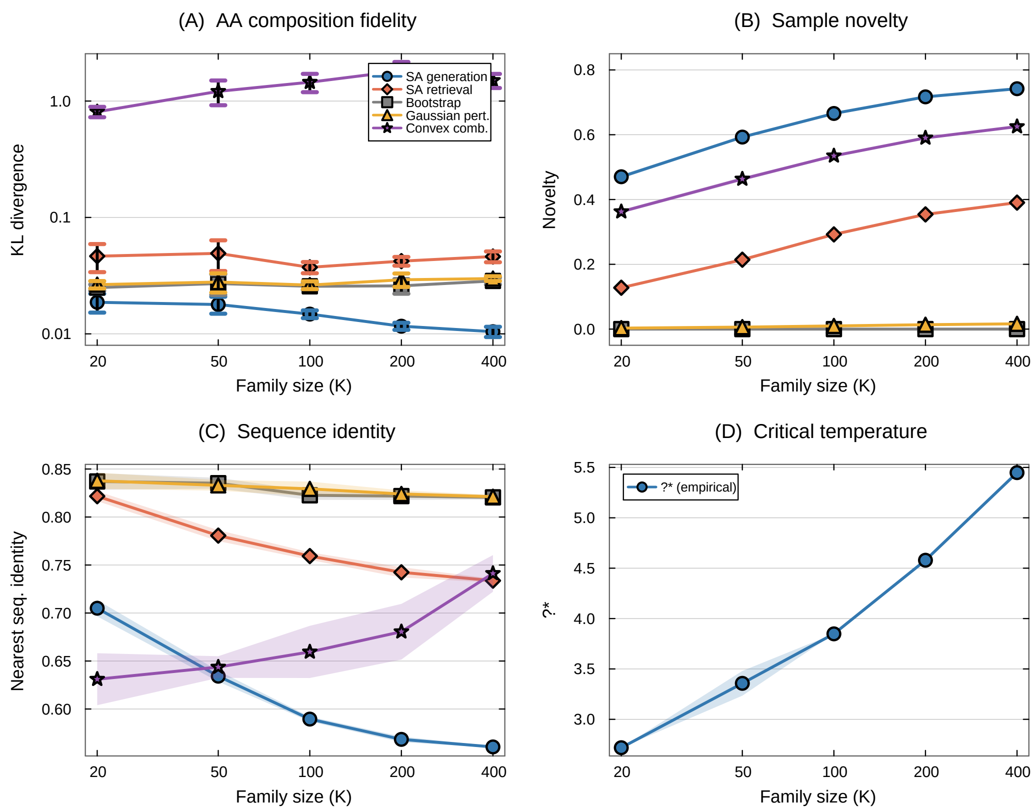

Figure 3: Scaling study: WW domain subsampled at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates. **(A)** KL divergence (log–log scale). SA generation maintains low KL across the entire range; convex combination degrades catastrophically. **(B)** Novelty increases with K for SA methods; baselines remain near zero. **(C)** Nearest sequence identity decreases with K, indicating the sampler explores further from stored patterns as more patterns shape the energy landscape. **(D)** The empirical critical temperature β^* increases with family size. Shaded regions and error bars show ± 1 SE.

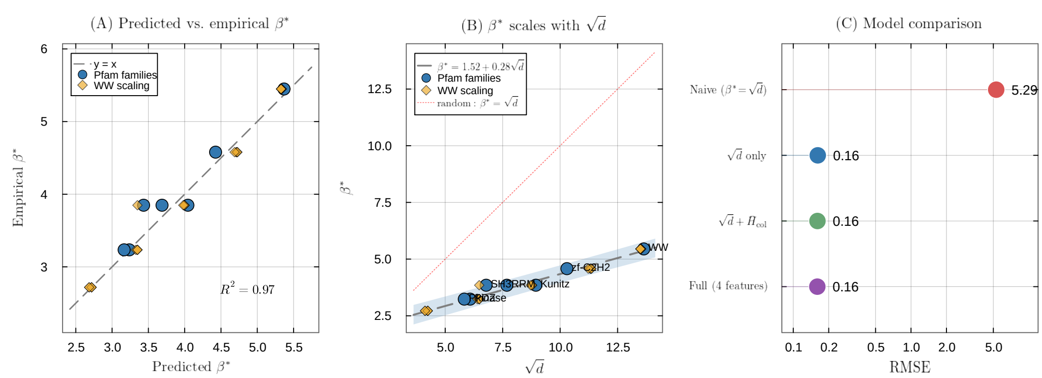

Figure 4: Predicting the critical temperature from MSA statistics. **(A)** Predicted vs. empirical β^* for the simple linear model $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (R^2=0.97, n=32). Blue circles: seven Pfam families; gold diamonds: WW scaling replicates ($K \in \{20, \ldots, 400\}$, five repeats each). **(B)** β^* as a function of $\sqrt{d}$, with the fitted regression (dashed) and the random-pattern prediction $\beta^* = \sqrt{d}$ (dotted red). The intercept offset and reduced slope reflect structured correlations in real protein families. **(C)** Model comparison (log-scale RMSE). The simple $\sqrt{d}$ model (RMSE=0.16) is 33× more accurate than the naive random-pattern prediction $\beta^* = \sqrt{d}$ (RMSE=5.29); adding MSA features provides negligible improvement.

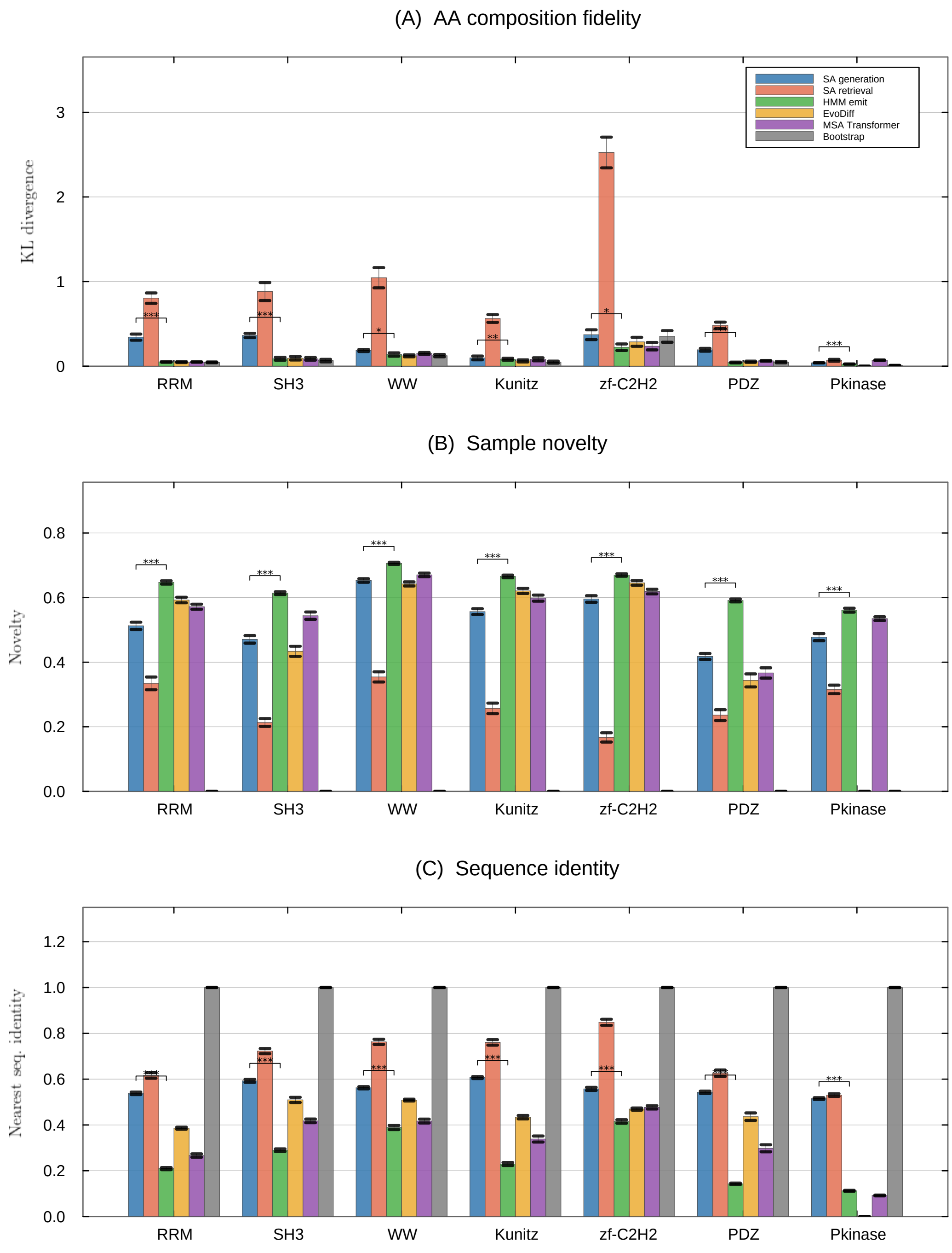

Figure 5: Head-to-head comparison of SA generation against learned baselines across seven Pfam families (six for EvoDiff, which could not complete Pkinase in feasible time). **(A)** KL divergence of amino acid composition. **(B)** Novelty in PCA space. **(C)** Nearest sequence identity. Significance brackets show Wilcoxon rank-sum tests comparing SA generation to HMM emit (the most direct comparison, as both use only the seed alignment): $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, n.s. = not significant. Error bars show ± 1 SE (30 chains $\times$ 5 samples).

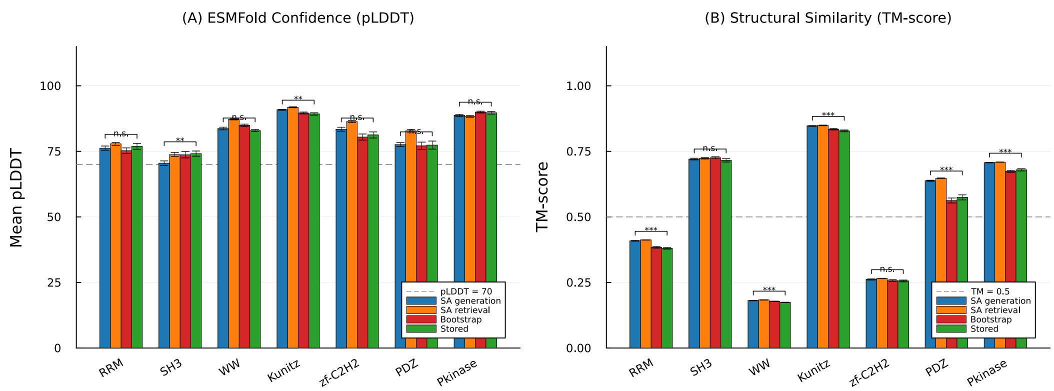

Figure 6: Structure validation of SA-generated sequences using ESMFold. **(A)** Mean pLDDT (folding confidence) across seven Pfam families for SA generation, SA retrieval, bootstrap, and stored (natural) sequences. The dashed line indicates the pLDDT=70 confidence threshold. **(B)** TM-score to a representative experimentally determined structure for each family. The dashed line indicates TM=0.5 (same-fold threshold). Significance brackets show Wilcoxon rank-sum tests comparing SA generation to stored sequences: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, n.s. = not significant. Error bars show ± 1 SE (n=50 sequences per condition).